

Harshit Agarwal

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EDUCATION:

University of Maryland (UMD), College Park, MD
PhD Physics

August 2026 – Present

Missouri University of Science and Technology (MST), Rolla, MO
B.S. Physics, Minor in Electrical Engineering

August 2024 – December 2025

GPA: 4.0/4.0

RESEARCH AND TECHNICAL EXPERIENCE:

Graduate Student Researcher

August 2026 - Present

University of Maryland, College Park

Research Assistant, Oak Ridge National Lab

January 2026 – July 2026

Supervised by **Dr. Benjamin Lawrie**

Project: *Spin based quantum sensing using NV centers and characterization of superconducting nanowires.*

- Mechanical and electrical integration of NV quantum sensors and superconducting nanowires into cryogenic testbeds.
- Performed **optical alignment** for widefield NV microscopy and **free-space coupling** of nanowires.
- Develop and implement **Python-driven automation pipelines** for sensor characterization and data acquisition.

Skills: Cryogenic testing and assembly, Superconducting Devices, Wire bonding, Microwave, DAQ, Lab Instrumentation.

Research Intern, National Institute of Standards and Technology (NIST), Boulder, CO

May 2025 - August 2025

Quantum Sensors Division

Project: *Measuring Polarization Response of Microwave Kinetic Inductance Detectors at sub-kelvin temperatures.*

- Developed a cryogenic polarization measurement system for MKIDs by integrating an infrared heater, optical filters, and membrane polarizers inside a dilution refrigerator.
- Designed a mechanical assembly using a UHV stepper motor on the 4K stage to rotate the optical stack, which eliminated systematic errors caused by external optical coupling.
- Modeled thermal loads to ensure cryostat stability and wrote Python scripts to automate control and data acquisition.

Skills: LABVIEW, Cryogenic testing and assembly, Lab Instrumentation, DAQ, Solidworks, Wire bonding, Ansys Electronics

Research Assistant, Dept. of Physics, MST

January 2025 – December 2025

Supervised by **Prof. Daniel Fischer**

Project: *Motorized Polarizing Michelson Interferometer for Laser Characterization.*

- Built and aligned a polarizing Michelson interferometer with a stepper motor-controlled mirror for path dependent interference measurements.
- Integrated an Arduino-based data acquisition system from photodetectors.
- Applied signal processing techniques using Python to extract laser frequency and coherence properties.

Skills: Lasers and Optics, Arduino, Command Line, Python, Fiber optics

Research Assistant, Dept. of Physics, MST

August 2024 – December 2025

Supervised by **Prof. Hyunsoo Kim**

Project 2: *Design, Fabrication and Measurement of a transmon qubit.*

- Designing a transmon qubit chip on Qiskit Metal; EM simulation using Ansys HFSS.
- Fabrication of chip using photolithography, electron beam lithography and thin-film deposition (sputtering).

Skills: Qiskit Metal, Ansys HFSS, Lithography, Microwave Engineering, VNA, Cryogenics, Superconducting Devices

Project 1 : *Measuring Properties of Topological Superconductor at cryogenic temperatures.*

- Designed and fabricated a self-oscillating LC circuit to measure the magnetic properties of quantum materials by tracking resonant frequency shifts.
- Optimized the circuit for cryogenic temperatures by characterizing transistor behavior across temperature gradients.
- Validated the circuit design using simulations and utilized KiCAD for PCB design and fabrication.

Skills: LABVIEW, Cryogenics, Lab Instrumentation, KiCAD, CNC Milling, PCB Fabrication and assembly, DAQ, Origin, Multisim

Research Assistant, LAMP Lab, Dept. of Mechanical and Aerospace Engineering, MST

August 2024 – Present

Project: [Implementation of Advanced Sensors and Machine Learning for Laser Aided Manufacturing](#).

- Integrating the sensors like Laser Disp. Sensor on hardware and software level to measure novel parameters.
- Application of Machine learning based on the data, for autonomous real-time adjustment of the factors (like Laser power, Powder Feed Rate, Layer height etc.)

Skills: High Power Lasers, Advanced Sensors, Machine Learning, G-Code, FPGA, Laser Displacement Sensor, CURA

REU, Case Western Reserve University

May 2024 – August 2024

Supervised by **Dr. Bill Yu**

Project: [Bio-inspired floating foundations for offshore wind turbines](#).

- Conducted extensive literature review of State-of-Art Wind Tech and existing challenges in the industry.
- Designed bio-inspired models to tackle instability, manufacturing and fatigue and 3-D printed for testing.
- Fabricated and performed various experimental tests to check stability and fatigue resilience.

Skills: Python, Arduino, Sensors, 3D Modelling, Hydraulic Press Fatigue Analysis, 3D Printing, COMSOL

Research Assistant, Centre of Nanoscience, UMSL

December 2023 – January 2024

Supervised by **Prof. Phil Fraundorf**

Project: [Analyzing the diffusive properties of Carbon Nano Tubes](#).

- Heated CNTs over 3500K in vacuum oven, analyzed the sample using SEM.
- Designed and conducted experiment to calculate resistivity of CNTs at ultra-high temperatures.

Skills: SEM, Vacuum pump, High Temperature oven, thermal probes

Observational Assistant, Dept. of Physics and Astronomy, UMSL

October 2023 – November 2023

Supervised by **Prof. Erika Gibb**

Project: [Investigation of volatile components on Comets' surface](#).

- Observed comets by remotely operating telescopes across the country.
- Recorded and analyzed data (IR Spectroscopy), and stacked images from the observations.

Skills: Remote Controlling telescopes, IR Spectroscopy, Python, Stacking, Data processing

Student Design Teams (at SDELC, MST)

August 2024 – May 2025

- **MSAT:** Flatbed Integration and Testing of Satellite.
- **Mars Rover:** Hardware level noise reduction for Raman Spectroscopy using LRC filters.
- **Multicopter:** Electronics layout for VTOL.

Upcoming Publications:

H. Agarwal, O. Uralska, J. Billingsley, M. Yamilov, H. Kim. [AC magnetic measurements with a self-oscillating LC circuit and its application to university education](#). Am. J. Phys. 94, 312–320 (2026).

H. Agarwal, O. Mecklem, X. Yu. [“Bio-Inspired Floating Foundations for Offshore Wind Turbines.”](#)
(Submitted to Wind Energy and Engineering Research)

A. K. Vaskuri, **H. Agarwal**, J. E. Austermann, J. D. Wheeler, S. Duff, M. Vissers, J. Hubmayr, J. Ullom, **“Cryogenic Rotary System for Measuring Polarization Response of Superconducting Kinetic Inductance Detectors”**
(under review SPIE)

Conference Presentations:

“Cryogenic Polarization Measurement of Superconducting Quantum Sensors” ([link](#))

- Poster at Fermilab Quantum Symposium 2025
- Plenary talk at NIST Symposium 2025

“Development of an LC self-oscillator for magnetic measurements at cryogenic temperatures” ([link](#))

- Poster at Fuller Research Prize (MST) 2025
- Poster at APS March Meeting 2025

“Bio-Inspired Floating Foundations for Offshore Wind Turbines” ([link](#))

- Poster at APS March Meeting 2025
- Poster at APS Division of Fluid Dynamics Meeting 2024

Teaching Experience:

Quantum Technical Intern at EnLuz

September 2024 – November 2024

- Prepared educational videos on Quantum Information Science for Qiskit Fall Fest 2024.
- Hosted live office hours for doubt solving.

Tutor at UMSL (on Campus)

November 2023 – May 2024

- Tutoring University students in Physics and Mathematics.
- Developing new techniques and tailored analogies for better understanding.

Facilitator at St. Louis Science Center

January 2024 – March 2024

- Performing variety of live experiments on stage to engage audience.
- Engaged diverse audiences by tailoring explanations to their interests.
- Helping in ideation, building and maintaining artefacts with the in-house production team.

Honors, Awards and Fellowships:

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| • Fuller Research Prize in Physics | Missouri S&T (2025) |
| • OURE Research Fellowship | Missouri S&T (2025) |
| • NIST Summer Undergraduate Research Fellowship | (2025) |
| • American Physical Society Travel Award for APS Annual Meeting | (2025) |
| • College of Arts and Sciences Travel Award | Missouri S&T (2025) |
| • Fuller Undergraduate Scholarship in Physics | (2025) |
| • FYRE Research Fellowship | Missouri S&T (2024) |
| • Junior Alumni Scholarship Award | UMSL (2024) |
| • Induction to Sigma Pi Sigma (Physics Honor Society) | (2024) |
| • Millennium Fellow (United Nations). | (2023) |

Leadership:

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| • Founder and President of Quantum Computing Club at Missouri S&T | |
| • Vice President of Society of Physics Students (SPS) at Missouri S&T | (Spring 2025) |
| • Treasurer of Society of Physics Students (SPS) at Missouri S&T | (Fall 2024) |
| • Senator at Student Government Association University of Missouri - St. Louis | (Spring 2024) |
| • Secretary of Society of Physics Students (SPS) at University of Missouri - St. Louis | (Spring 2024) |
| • Vice President of Chemistry Club at University of Missouri - St. Louis | (Fall 2023, Spring 2024) |

Outreach and Engagement:

- **Founded Quantum Computing Club** at Missouri S&T
- **Demonstrated superconductor levitation** and quantum concepts at middle schools, high schools, and Science Fests.
- Hosted **student lab tours** for events such as National STEM Day.
- Designed the "**Optics**" **exam** (Division B and C) for the 2025 Regional Science Olympiad; proctored the on-site test and assisted in event organization.
- American Physical Society (APS) **Student Ambassador**.
- **Social Entrepreneurship** at Enactus DTU.

TECHNICAL SKILLS:

Programming:	Python, C/C++, Qiskit, MATLAB, LaTeX, G-Code, Data Analysis
Software:	COMSOL, MS Office, KiCAD, SolidWorks, LabView, Multisim, HFSS, KLayout
Hardware:	Microcontrollers, CNC Milling, Cryogenics, 3-D Printing, VNA, Vacuum Systems, PCB Design
Fabrication:	Optical Lithography, E-beam lithography, Thin-film deposition, Wire Bonding, Chip Packaging